A Personalized Real-Time Vocabulary Coaching Assistant for Second Language English Writers

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***Abstract*—The writing assistants powered by artificial intelligence are currently commonly employed in the stylistic refining of the text and grammatical error correction. However, most of the existing tools are focusing on automated rewrites or superficial corrections, which do not offer much learning to users who are learning English as a second language (ESL). This project is aimed at developing a real-time, personalized vocabulary coaching assistant, which can only be used as a text-proofreading aid, but not as a text-generating system. In live writing, the system identifies non-optimal lexical options, provides contextually applicable alternatives and provides brief explanations and examples of each proposal. The suggested assistant aims at reducing the distance between correction of automated writing and effective acquisition of the second language through emphasis on transparency, user learning and vocabulary development.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

The recent breakthroughs in natural language processing have resulted in the wide usage of AI-based writing assistants across scholars, work, and personal settings. Grammar checkers, paraphrasing systems and other tools can quickly make a person more fluent and correct, but they are usually black-box systems that do not encourage a deeper understanding of the language. This becomes a serious constraint to those studying English as a second language: errors are being rectified, and vocabulary development and in-depth understanding of the language is often missed.

The empirical research in second language acquisition focuses on the idea that vocabulary acquisition is a key factor that defines the writing competence, reading comprehension, and long-term language recovery. This is notwithstanding that the majority of commercial writing aids concentrate on restructuring sentences, modulating tone, or fully writing a piece of writing instead of focused training in vocabulary. Additionally, text generation systems that create text directly pose a threat to agency on the part of learners and are likely to foster passive dependency on technologies instead of active capacity building.

This project would fill these gaps by suggesting a vocabulary coaching helper which is real time and only interferes on the level of lexical choice. The system identifies any generic or imprecise use of words by the user as he or she writes, and suggest better quality alternatives that seem to fit in the context of the sentence and the level of proficiency the user has. Most importantly, there is a concise explanation on why the alternative is better and a sample sentence on how the alternative would be used. This design is such that corrections can be read, learnable, and taken up at once.

The proposed assistant, unlike the current tools, will not produce original content, paraphrase whole sentences, or distort the intention of the author. Rather, it plays the role of a smart tutor that is part of the writing process. This system will assist in vocabulary learning with the integration of real-time feedback, personalization, and clear pedagogical explanations without losing user control of the writing process.

II. Problem Statement

The principle problem that the given project is trying to address is that existing AI writing aids in second language English learners do not allow providing real-time, pedagogically-based feedback on vocabulary. The current tooling either focuses on grammatical precision or provides the general suggestions without clarifying the linguistic reasoning behind the change in vocabulary. Consequently, users can be short-term better writers but fail to increase their vocabulary and become better writers independently. The hypothesis of this work is that a personalized, explanation-driven vocabulary coaching system can improve lexical sophistication and long-term learning outcomes for ESL writers without relying on auto- mated text generation.

III. Motivation

The knowledge of vocabulary is the fundamental aspect of good written communication, especially when it comes to non-native speakers of English in a school or workplace. The problem with learners is that learners do not have ideas, but are unable to find the right words with the right context to convey these ideas. The so- lutions to this challenge are of practical value as they can lead to better academic results, greater confidence in writing, and minimized reliance on automated rewriting programs. In a broader scope, the study will help in building human-oriented artificial intelligence systems that can focus on learning and transparency rather than automation. This project meets the criteria of ethical AI and educational best practices by creating an assistant that has the ability to explain its actions, as well as, uphold the agency of its users. Practical relevance of the proposed approach to the language education platforms, productivity tools in the workplace and writing technologies oriented on accessibility are also suggested.

IV. Background and Related Work

The Author examines the extent to which various aspects of vocabulary knowledge, in general, vocabulary size and vocabulary depth, in particular are predictors of overall language proficiency in the four skills namely reading, writing, listening, and speaking, with vocabulary being a significant predictor of performance in all of them. Milton [1] has given empirical evidence on the importance of vocabulary to second language learner interventions to support the notion that the focus on lexical development is not an peripheral issue but rather a central pathway to enhanced communicative competence. The significance of Milton (2013) is that it shows vocabulary measures can explain a large proportion of the variance in proficiency scores, often more than other linguistic sub components, which means that improving learners' lexical breadth and depth should directly improve their writing and other skills. A major drawback of the research is that the vocabulary size and depth were determined by prescribed standardized tests which might not be a good measure of what learners know about lexical productive and contextual knowledge in actual communicative contexts. As well, the results are derived on a specific group of learners and a specific set of tasks, thus it is unclear how far the findings can be extended to other levels of proficiency, tasks, and types of assessment.

Ranalli, Link, and Chukharev-Hudilainen discuss the potential to use an automated writing evaluation (AWE) system as a way to conduct formative assessment in writing in the second language (L2), and the research question is whether the feedback provided by an AWE system is accurate and useful to learners and teachers in the teaching process. The article [2] has a crucial concern , as it is one of the most important lines of work concerning AI-based writing support tools that are almost always based on offering corrective feedback on grammar, mechanics, and other surface characteristics, demonstrating both their advantages (improved accuracy, more revisions, increased fluency) and their disadvantages. The article gives us the understanding to work more on the more explanation-rich, vocabulary-oriented, real-time assistance that specifically addresses the gaps that AWE tools do not fully address.The study however has certain limitations in that it was confined in one AWE (Criterion) and a particular learner group and thus it limits the generalization of the results of the study on other tools and genres and instructional settings. Moreover, since accuracy differed significantly among the different types of errors and the system was primarily focused on surface-level features, the study does not comprehensively represent how AWE may be useful in higher-level features of L2 writing including global structure, knowledge of the audience, and content elaboration.

Koltovskaia (2020) examines the behavioral, cognitive, and affective reactions of ESL university students when using the Grammarly automated corrective writing feedback to revise their writing. The research indicates that automated feedback does not automatically motivate students to accept or reject the feedback, rather, their engagement is determined by their level of trust in the tool, their comprehension of the feedback, their motivation, and their previous experiences with both English and technology. This article [3] points out one of the fundamental weaknesses of the current automated feedback technologies, which are that as much as feedback is present, a learner might not intelligibly process or learn it, particularly when it is opaque and overwhelming. Its relevance to the proposed vocabulary coaching assistant lies in the fact that it highlights the importance of creating feedback that is clear, abundant in explanations and pedagogical friendliness, so that learners can actively respond to suggestions and transform them into the real vocabulary acquisition as opposed to the mere surface correction.The study has its shortcomings however in its small multiple case-study sample, comprising of the two ESL students which used a single tool (Grammarly), which limits the application of the results to the large L2 learner populations and other automated feedback systems. Additionally, since the data will belong to a particular course and situation, the study will not be able to comprehensively represent how various levels of proficiency, tasks, or modes of instruction may influence learners behavioral, cognitive, and affective interactions with automated feedback.

The authors Huang et al. (2023) use the opportunities of ChatGPT to facilitate second language learning in the article, discussing its possible purposes as a partner in the conversation, a provider of feedback, and a generator of learning resources, and the perceptions of its usefulness and limitations students have. This paper [4] is a new dedicated study aimed at contextualizing ChatGPT in the context of L2 education that reports not only the advantages (e.g., rich input, personalized interaction, high availability) but also the disadvantages (e.g., accuracy, over-reliance, and alignment with pedagogical goals). Its relevance to the proposed vocabulary coaching assistant is in the contrast it assists to provide: unlike ChatGPT, which is an impressive general-purpose generator that can assume much of the writing process, the assistant in this project is specifically limited to be non-generative and instead be explanation-based vocabulary assistance, to harness AI to learn the language and to develop it actively instead of passively reading a text. The research is, however, constrained by the fact that it used self-reported perceptions, on a particular group of learners in a certain institutional environment, which limits the extent to which the study findings can be extrapolated onto other groups and contexts of L2. Moreover, since it is mostly exploratory and limited to the short-term classroom application, it does not present meaningful evidence concerning the long-term outcome of the ChatGPT on the real language development or the long-term independence of the learners.

Warschauer and Tong (2023) address the way generative

AI systems like ChatGPT are transforming the teaching of writing, stating that these systems will allow creating new opportunities to support students and raising important questions about authorship, assessment, and equity in the writing classroom. The paper [5] gives a conceptual framework to the understanding of the potential use of AI-generated text as supplementary, but not substitutive, of the core writing procedures of planning, drafting, and revising, and underlines the importance of explicit AI literacy teaching students how to perceive, prompt, evaluate, and own the AI output into their own writing. Its value to the proposed vocabulary coaching assistant is that it allows to place the project within the line of becoming an example of AI-as-support, rather than AI-as-replacement: the assistant does not produce full texts, but rather provides limited, interpretable lexical assistance that supports, rather than replaces, the writing and learning of students. Nevertheless, the article itself is more conceptual and illustrative, thus, failing to give empirical data on the actual impact of generative AI on the writing development or classroom practices of students in a variety of settings. It is also limited in the number of educational contexts and examples of tools and therefore its framework cannot be generalized which leaves undefined how challenges of equity, access and assessment will manifest in other institutions and learners.

The study by Shintani (2016) [6] explores the effectiveness of computer-assisted pronunciation feedback in helping second language learners to achieve higher pronunciation accuracy and the results are compared between the learners who get technology-based feedback and those who get more traditional conditions. The research problem is the impact of instantaneous, intensive feedback on the verbal production of learners, which can have them given attention to problematic sounds, and demonstrate that the problematic performance on L2 pronunciation caused by dedicated, computer-based feedback can improve over time. The paper shows that a fine-grained feedback involving a specific sub skill (in our case, pronunciation) with the help of computers may be pedagogically effective, which supports the larger notion that technology can be developed not only to correct language, but to facilitate learning of a particular element of competence. It is important to the proposed vocabulary coaching assistant in the sense that it provides a parallel: in the same way as segmental and suprasegmental accuracy can be enhanced through pronunciation-based feedback system, a well-thought-out assistant that is vocabulary-focused can use real-time explicit feedback on lexical choice to both develop more profound lexical knowledge and enhance more accurate and sophisticated use of words in L2 writing.The study, though, is restricted because it uses a small learner group, target sounds, and instructional situation, which limits the extent of generalization of the results to other L2 groups and to other features of pronunciation. It also looks at comparatively short term benefits hence it offers less data on the viability of the enhancements made by computer assisted pronunciation feedback in the long term.

Altogether, the literature review highlights the significance of vocabulary training and the opportunities of AI-based feedback systems. Nevertheless, none of the currently available strategies are able to integrate real time lexical intervention, customized scaffolding, overt pedagogical exposition, and system imposed non-generative constraints in an integrated framework. This is where the proposed vocabulary coaching assistant is motivated.

V. Research on Dataset

The section contains the datasets used by different author for various purpose. deciding on the final dataset corpus needs more research and fine tuning , but you can have a clear idea of what is the dataset that can be actually used in the project.

A. TOEFL11 (ETS Corpus of Non-Native Written English)

A large, proficiency-marked corpus of non-native English essays, that would be good at extracting lexical characteristics that were conditioned by the level of the learner and at assessing whether your vocabulary coaching assistant is causing students to shift towards the patterns of the higher-scoring texts. [7]

B. FCE (First Certificate in English) Dataset

The Cambridge First Certificate in English (FCE) dataset is used as the source of ESL data. The corpus is a subset of the Cambridge Learner Corpus (CLC) and contains English essays written by upper-intermediate level learners of English for the FCE examination.

C. ICNALE (International Corpus Network of Asian Learners of English)

The ICNALE (International Corpus Network of Asian Learners of English) is an international learner corpus developed by Dr. Shin Ishikawa, Kobe University, Japan. The ICNALE includes more than 15,000 topic-controlled speeches and essays produced by more than 5,800 college students (incl. grad students) in ten countries/ regions in Asia (China, Hong Kong, Indonesia, Japan, Korea, Pakistan, the Philippines, Singapore/ Malaysia, Taiwan, and Thailand) as well as English native speakers. The ICNALE WEP, a recent addition, includes data from the students in Bangladesh, Brunei, Cambodia, India, Laos, Malaysia, Myanmar, Nepal, and Vietnam. The ICNALE, which currently comprises five data modules: Spoken Monologue, Spoken Dialogue, Written Essays, Written Essays Plus, and Edited Essays, has become one of the largest learner corpora publicly available. [8]

More research has to be done in order to decide on the final dataset/corpus to work on.

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